

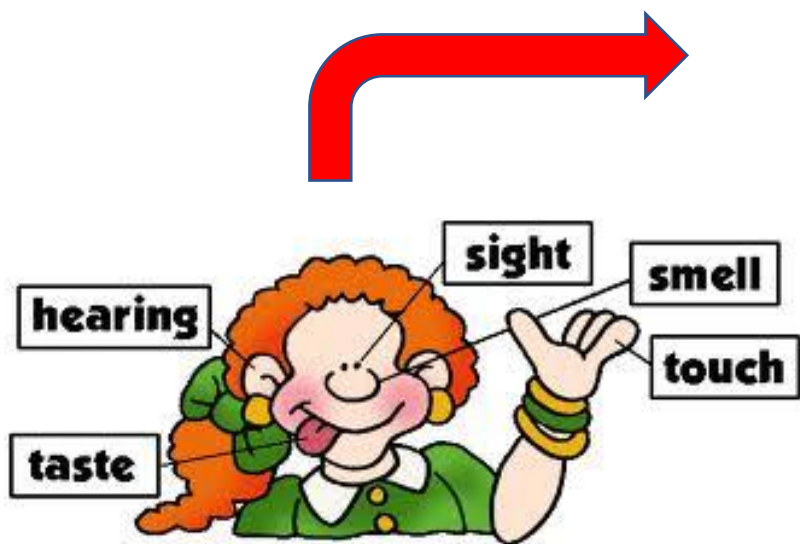
多模态信息处理

张长水

清华大学 自动化系

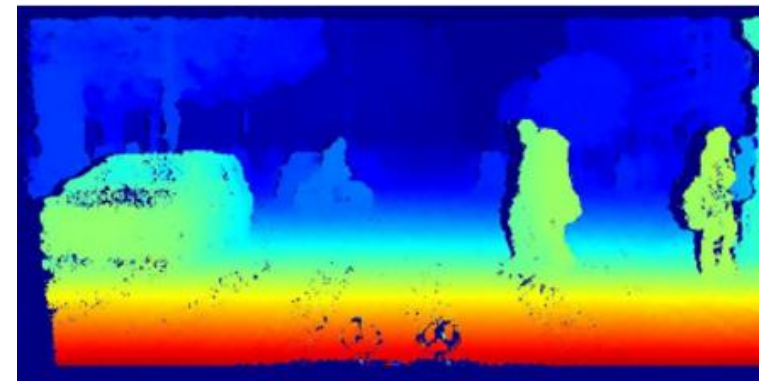
智能体感知世界

- 多模态信息：视觉、听觉、触觉、味觉、嗅觉



多模态数据

- 视觉、听觉
- 语言：超模态
- 触觉、嗅觉
- 红外图像、X光图像、核磁共振图像、深度图像
- 脑电信号、皮电信号、手机位置信号



小史

- 视频: gaga
- 声音: baba
- 效果: dada
- 听觉和视觉的相互作用

The McGurk Effect (1976)



[Hearing lips and seeing voices – Nature](#)

小史

- 语音识别：加入说话人的口型信息
- 多媒体计算
 - 视频镜头切换检测
 - 视频解译
 - 视频摘要
 - 视频检索
 -
- 深度学习时代：多模态学习
 - 采用深度学习方法研究多模态问题

多模态学习任务：语言引导的任务

- 语言指导的图像编辑



•Changming Xiao, Qi Yang, Xiaoqiang Xu, Jianwei Zhang, Feng Zhou, Changshui Zhang. **Where You Edit is What You Get: Text-Guided Image Editing with Region-Based Attention.** *Pattern Recognition*. 2023.



Rosy Cheeks

Bushy Eyebrows

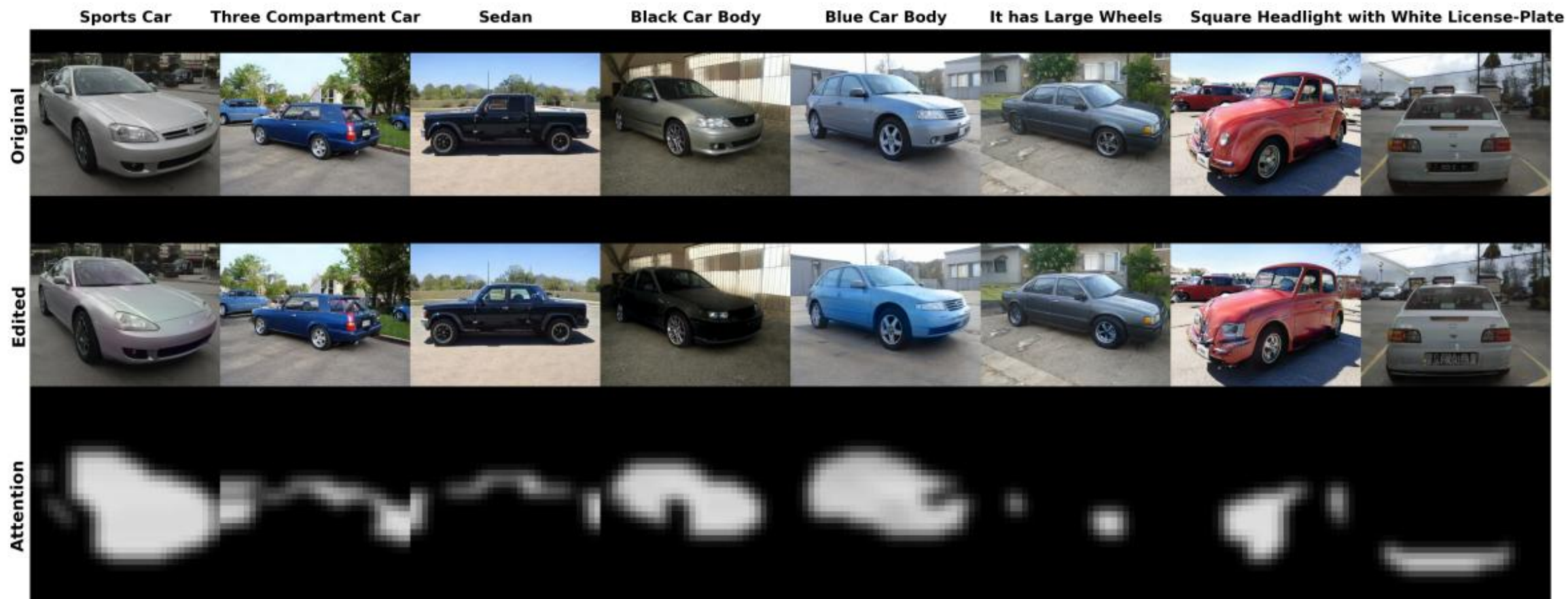
Pink Lipsticks

Mouths are Slightly Open

Blonde Hair

Blonde Hair

•Changming Xiao, Qi Yang, Xiaoqiang Xu, Jianwei Zhang, Feng Zhou, Changshui Zhang. **Where You Edit is What You Get: Text-Guided Image Editing with Region-Based Attention.** *Pattern Recognition*. 2023.



•Changming Xiao, Qi Yang, Xiaoqiang Xu, Jianwei Zhang, Feng Zhou, Changshui Zhang. **Where You Edit is What You Get: Text-Guided Image Editing with Region-Based Attention.** *Pattern Recognition*. 2023.

多模态学习任务：语言引导的任务

- 语言指导的图像/视频生成



fine watercolor painting portrait of beautiful girl holding a lazy cat, neat, elegant, by ilya kuvshinov, by ben quilty, by hikari shimoda

<https://mp.weixin.qq.com/s/Q4ZYjUxt22Jsx2W2179C8Q>



“一艘青花瓷质感的战舰”

文心一格 (<https://yige.baidu.com/>)

多模态学习任务：语言引导的任务

- 语言指导的视频分析

Sentence query: The second girl joins her again and they finish dancing



man in green shirt standing
man in yellow shirt jumping over a man



多模态学习任务：语言引导的任务

- 语言指导的视频分析



00:00:03,576 --> 00:00:05,697
Gavin Mitchell's office.
Rachel Green's office.

00:00:05,870 --> 00:00:07,409
Give me that phone.

00:00:08,873 --> 00:00:12,293
Hello, this is Rachel Green.
How can I help you?

00:00:12,460 --> 00:00:17,629
Uh-huh. Okay, then.
I'll pass you back to your son.

00:00:18,800 --> 00:00:21,639
Hey, Mom. No, that's just my
secretary.

(positive) The woman becomes upset when the man answers the phone because he pretends it is his own office.

(negative) The woman becomes upset when the man answers the phone because she is expecting a phone call from her mom.

Inferring reasons

(positive) The woman realizes it is the man's mother who is calling and she passes the phone back to the man.

(negative) The man realizes it is the woman's mother who is calling and he passes the phone back to the woman.

Identifying characters

(positive) The phone rings, a man picks it up, and a woman slams her hand on the desk and demands the man give her the phone.

(negative) The two people that the man in the glasses is talking to need to be briefed on something.

Global video understanding

多模态学习任务：语言引导的任务

- 视觉-语言导航
- 数据集：Room-to-room (R2R)



Leave the bedroom, and enter the kitchen. Walk forward, and take a left at the couch. Stop in front of the window.

<https://bringmeaspoon.org/>

Leaderboard <https://evalai.cloudcv.org/web/challenges/challenge-page/97/leaderboard/270>

多模态学习任务

- 用语音描述图像、音视频



a man sitting at a table
using a laptop



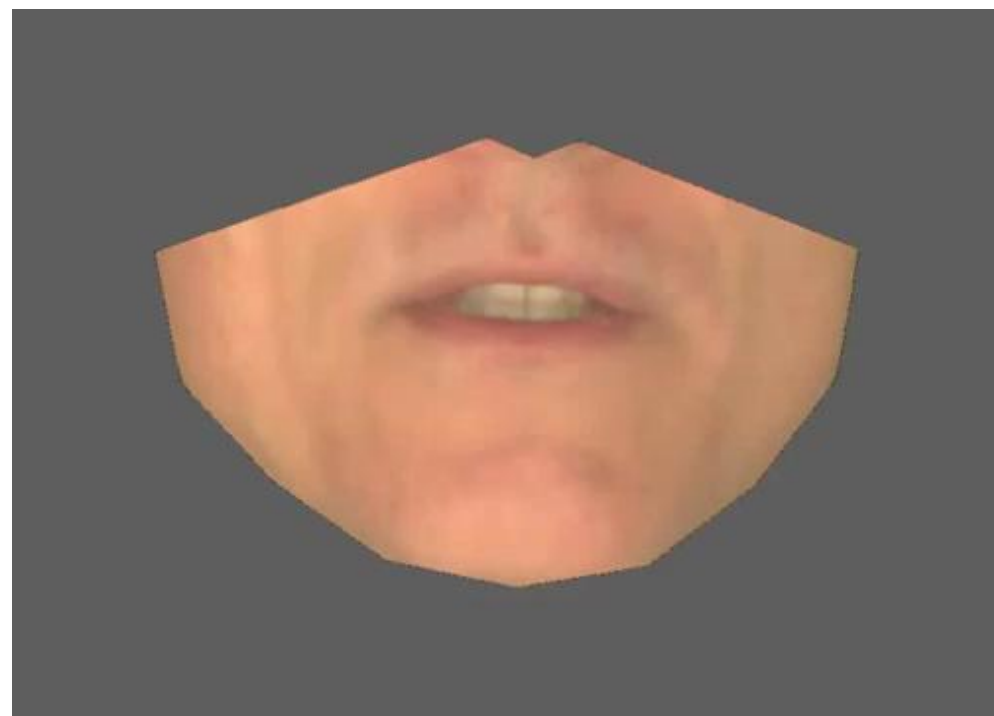
a white plate topped with
a half eaten sandwich



a group of people walking
in the rain with umbrellas

多模态学习任务

- 根据音频生成图像/视频
- 二维
- 三维

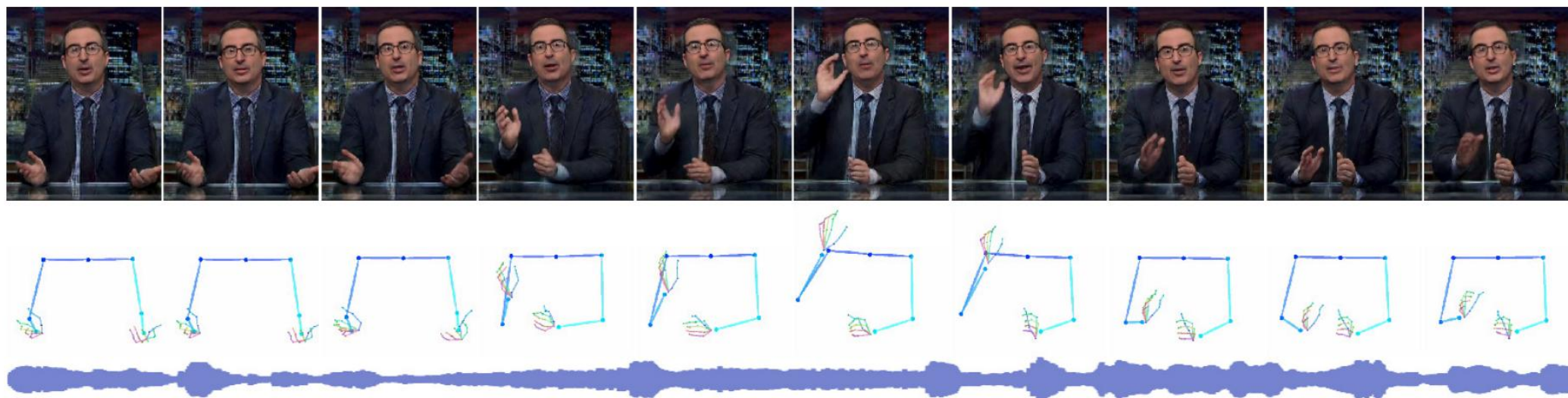


翻译

最近我去了很多地方

多模态学习任务

- 根据音频生成图像/视频



多模态学习任务

- 根据音频生成图像/视频
- 根据文本生成视频



•Runpeng Cui, Zhong Cao, Weishen Pan, Changshui Zhang, Jianqiang Wang. **Deep Gesture Video Generation with Learning on Regions of Interest.** *IEEE Transactions on Multimedia.* 2020.

•Runpeng Cui, Hu Liu, Changshui Zhang. **A Deep Neural Framework for Continuous Sign Language Recognition by Iterative Training.** *IEEE Transactions on Multimedia,* 2019

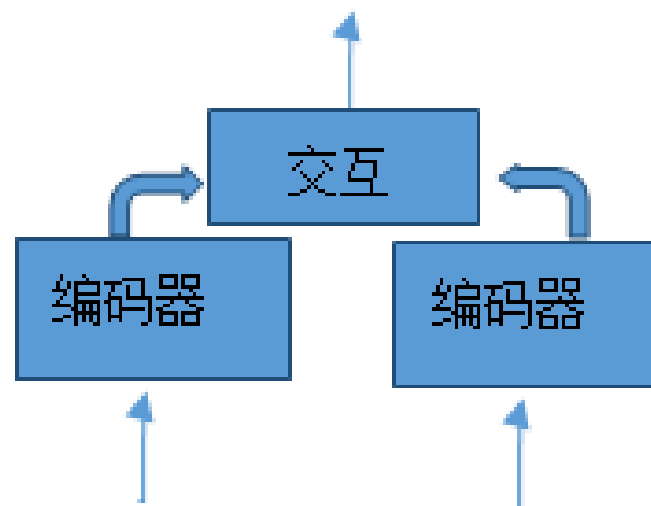
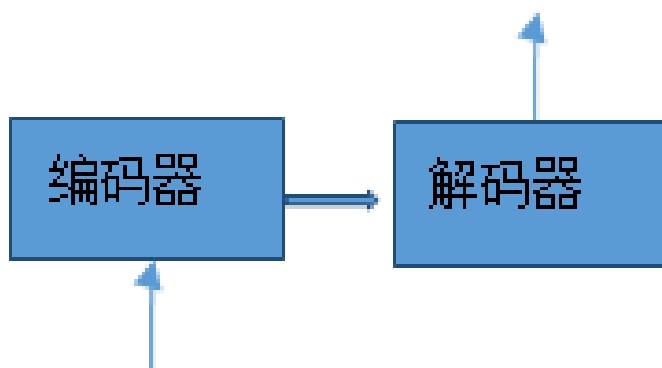
多模态学习任务

- 基于图像/视频的对话问答



方法

- 两大类方法



关键问题： 对齐

- 显示对齐

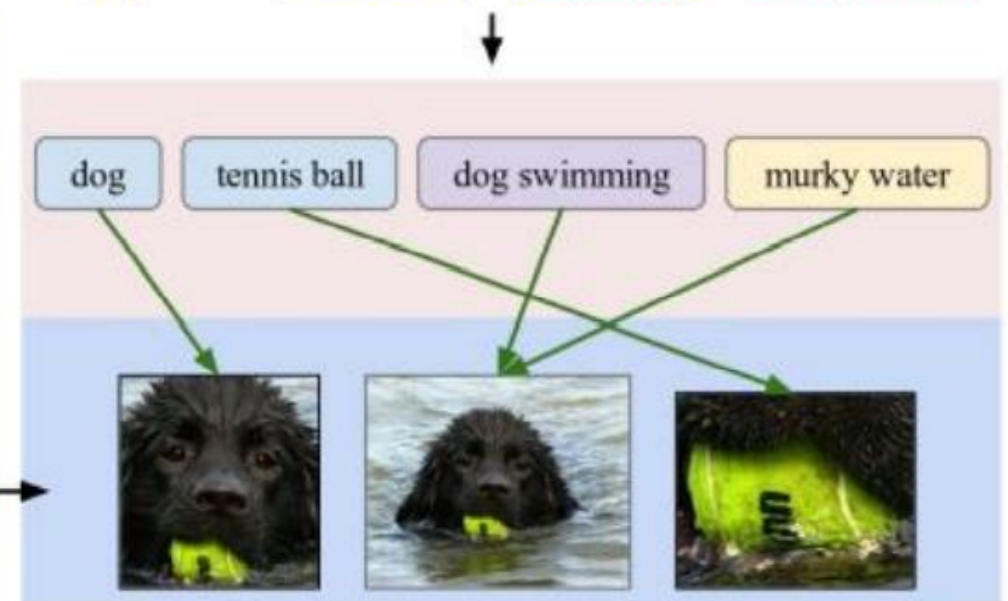


关键问题： 对齐

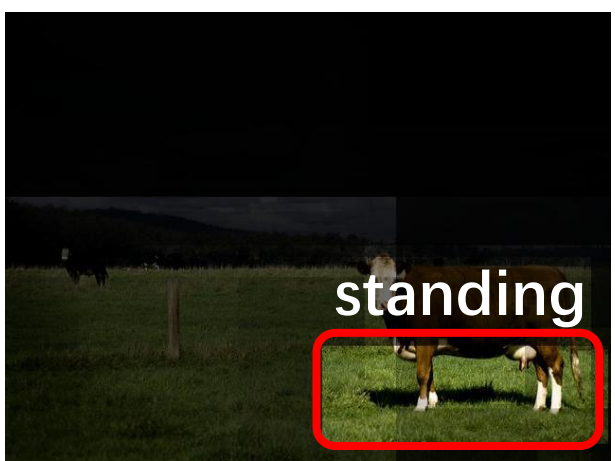
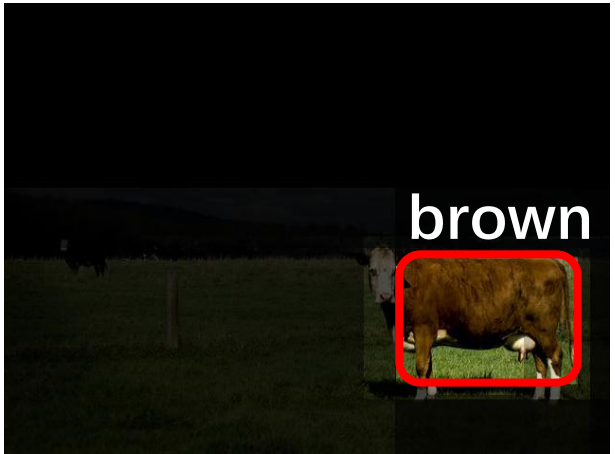
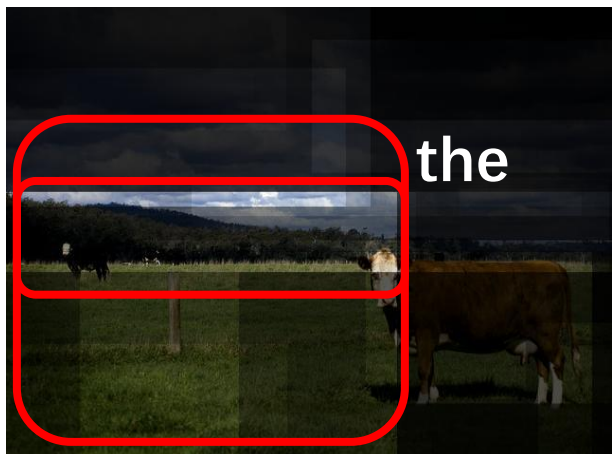
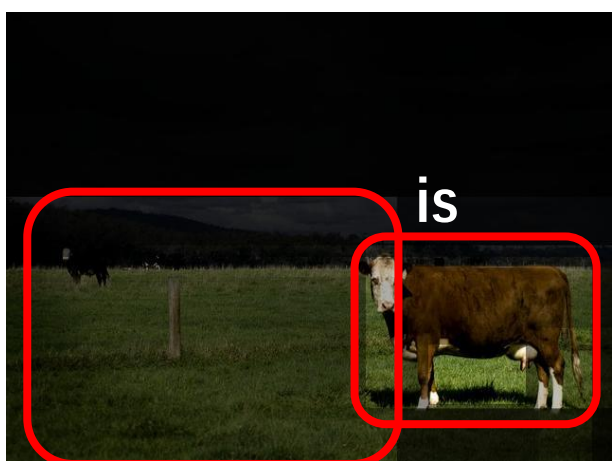
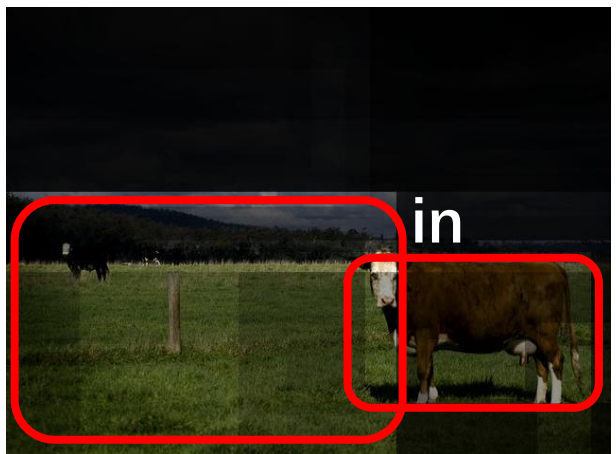
- 隐式对齐



“A dog with a tennis ball is swimming in murky water”



Karpathy et al., Deep Fragment Embeddings for Bidirectional Image Sentence Mapping,
<https://arxiv.org/pdf/1406.5679.pdf>



关键问题： 对齐

- 隐式对齐

fry



branch



关键问题： 对齐

- 隐式对齐

wooden



red



•Kun Fu, Junqi Jin, Runpeng Cui, Fei Sha, Changshui Zhang. **Aligning where to see and what to tell: image captioning with region-based attention and scene-specific contexts.** *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*. 2017

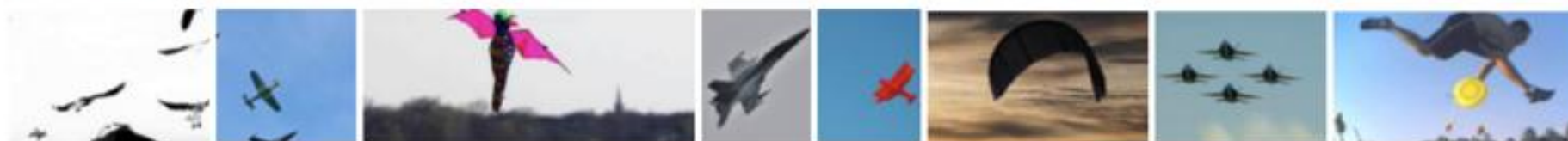
关键问题： 对齐

- 隐式对齐

bite



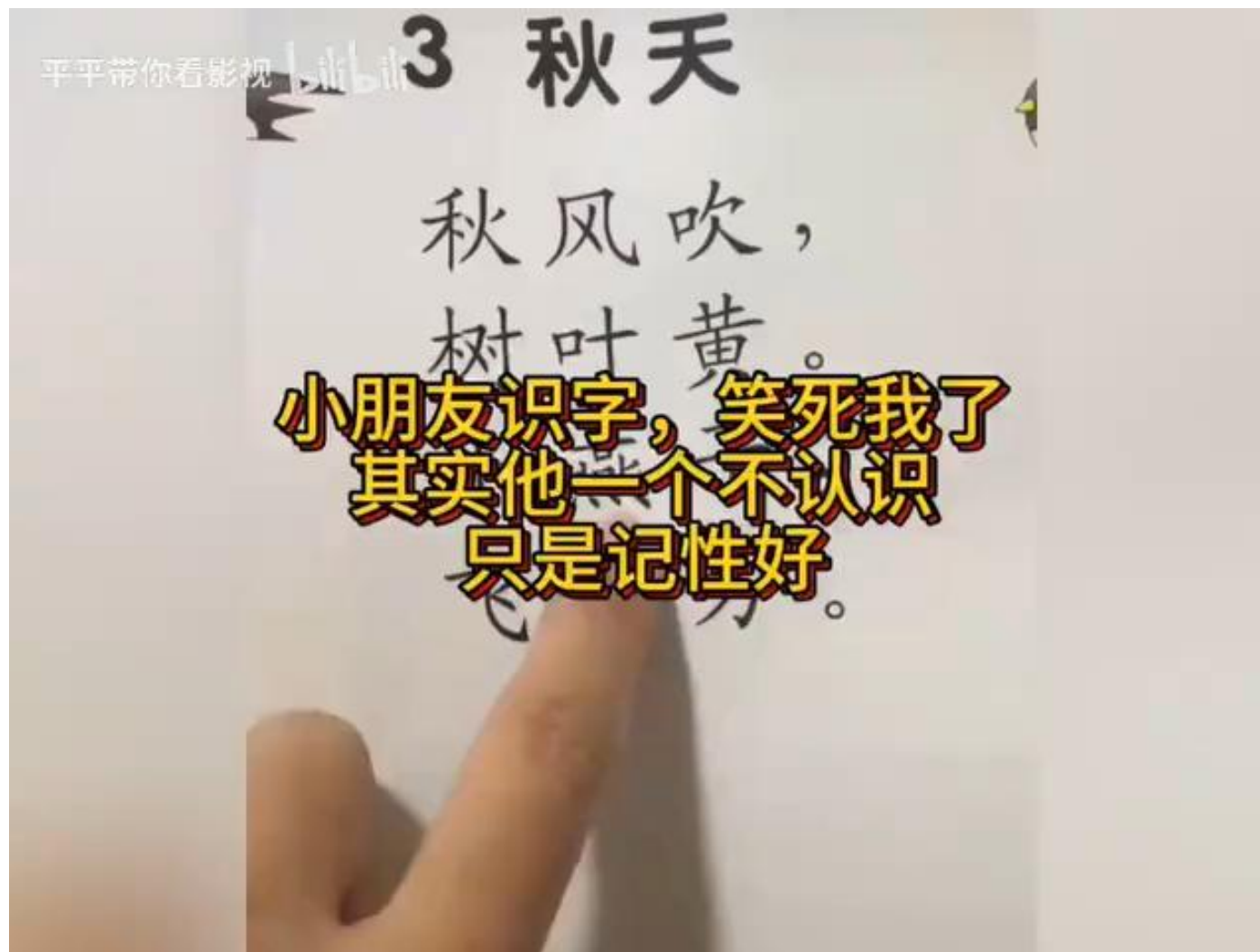
fly



•Kun Fu, Junqi Jin, Runpeng Cui, Fei Sha, Changshui Zhang. **Aligning where to see and what to tell: image captioning with region-based attention and scene-specific contexts.** *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*. 2017

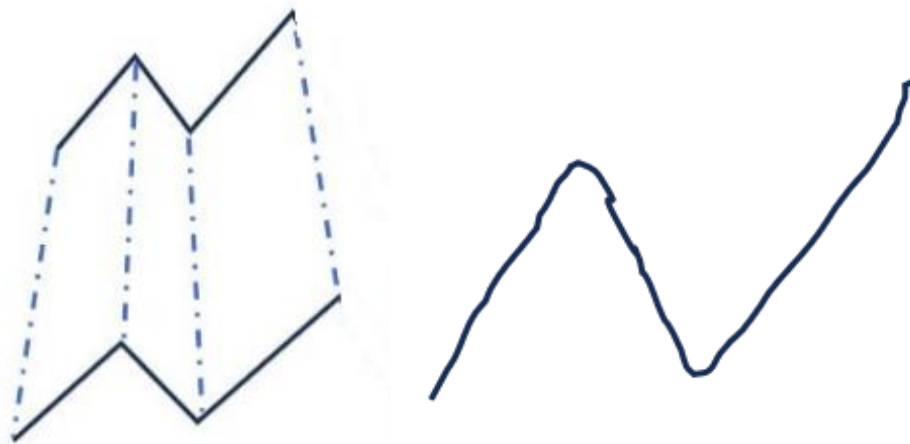
关键问题： 对齐

- 对齐



关键问题： 对齐

- 折线1={S,L,L,L,P,L,L,V,L,L,L,L,E}
- 折线2={S,L,L,L,L,P,L,L,L,V,L,L,L,L,L,E}



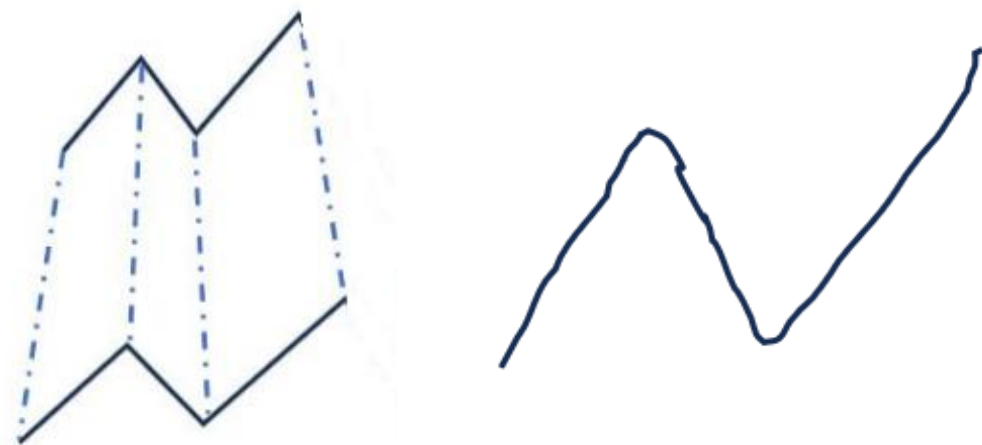
曲线 1={0.01, -0.02, 0.01, 9, 0.01, -0.01, -10, 0.01, -0.01, 0.02, 0.03}

曲线 2={0.01, -0.02, 0.01, 0.02, 9.5, 0.01, -0.01, 0.01, -10, 0.01, -0.01, 0.02, -0.01, 0.03}

误差为:

$$\begin{aligned} & (0.01 - 0.01)^2 + (-0.02 + 0.02)^2 + (0.01 - 0.01)^2 + (0.01 - 0.02)^2 + (9 - 9.5)^2 + \\ & (0.01 - 0.01)^2 + (-0.01 + 0.01)^2 + (-0.01 - 0.01)^2 + (-10 + 10)^2 + \\ & (0.01 - 0.01)^2 + (-0.01 + 0.01)^2 + (0.02 - 0.02)^2 + (0.03 + 0.01)^2 + (0.03 - 0.03)^2 \\ & = 0.2521 \end{aligned}$$

关键问题： 对齐



曲线 1={0.01, -0.02, 0.01, 9, 0.01, -0.01, -10, 0.01, -0.01, 0.02, 0.03}

描述 1={L, L, L, P, L, L, V, L, L, L, L}

关键问题： 对齐

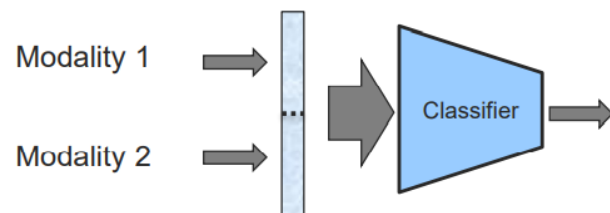
- 语言和图像的对齐： grounding(落地)
- 扶桑花
- 杜鹃花



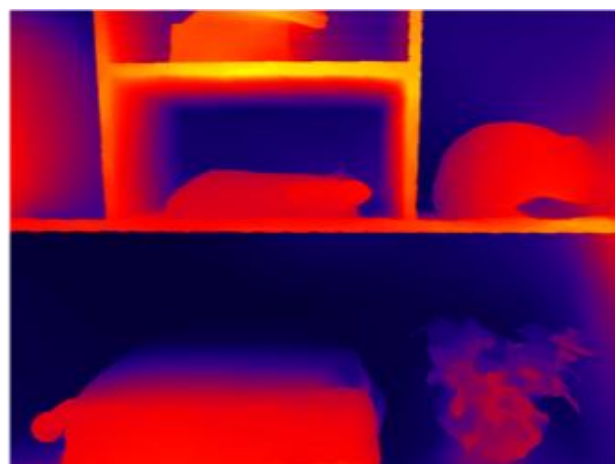
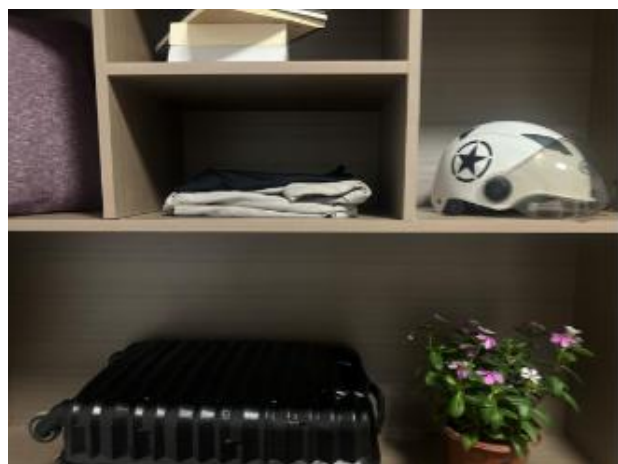
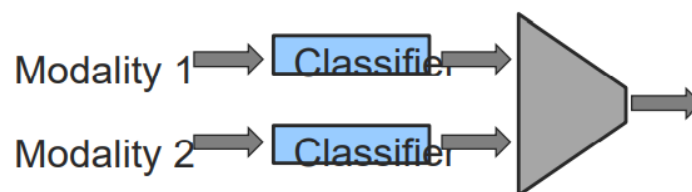
关键问题：融合

- 预测前融合

1) Early Fusion



2) Late Fusion



关键问题：融合

- 预测后融合

利用图像信息给出的类别预测结果如下：

物体A：茶杯0.6、猫0.4

物体B：茶杯0.1、猫0.9

利用声音信息给出的类别预测结果如下：

物体A：茶杯0.9、猫0.1

物体B：茶杯0.1、猫0.9

关键问题：融合

- 预测后融合

利用图像信息给出的类别预测结果如下：

物体A：茶杯0.6、猫0.4

物体B：茶杯0.1、猫0.9

利用声音信息给出的类别预测结果如下：

物体A：茶杯0.9、猫0.1

物体B：茶杯0.1、猫0.9

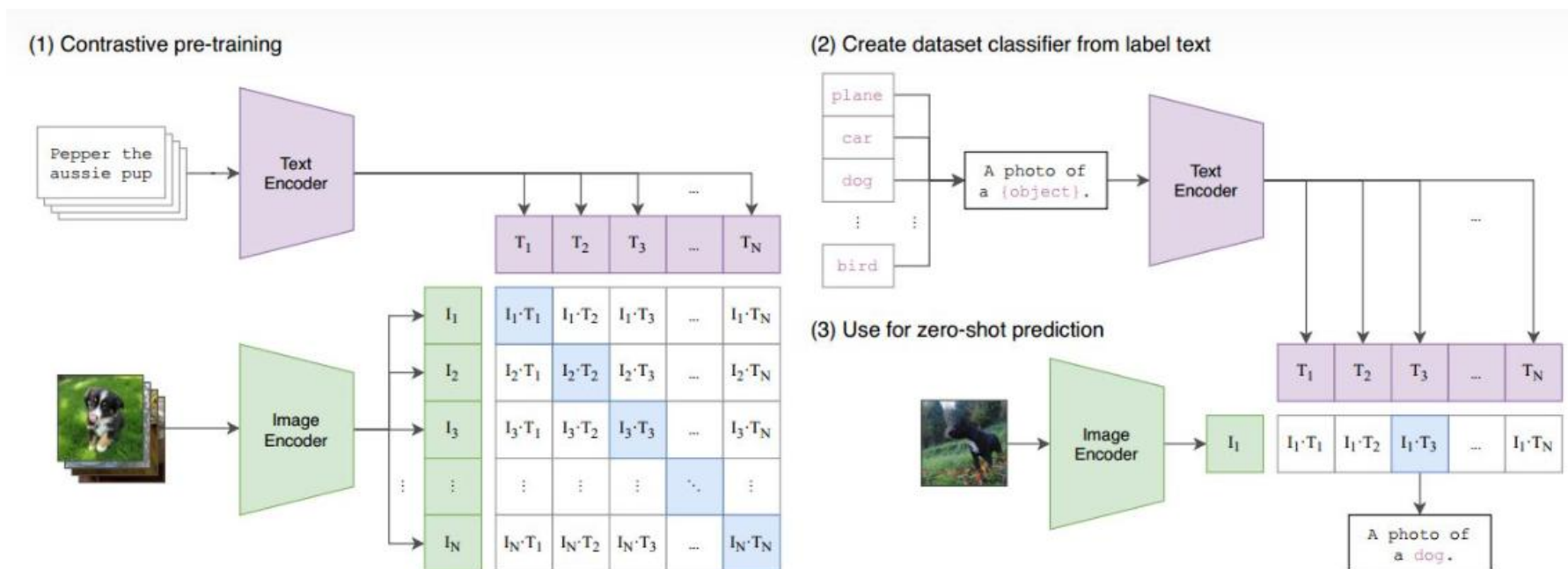
通过投票可以得到如下预测结果：

物体A：茶杯 $(0.6+0.9)/2=0.75$ 、猫 $(0.4+0.1)/2=0.25$

物体B：茶杯 $(0.1+0.1)/2=0.1$ 、猫 $(0.9+0.9)/2=0.9$

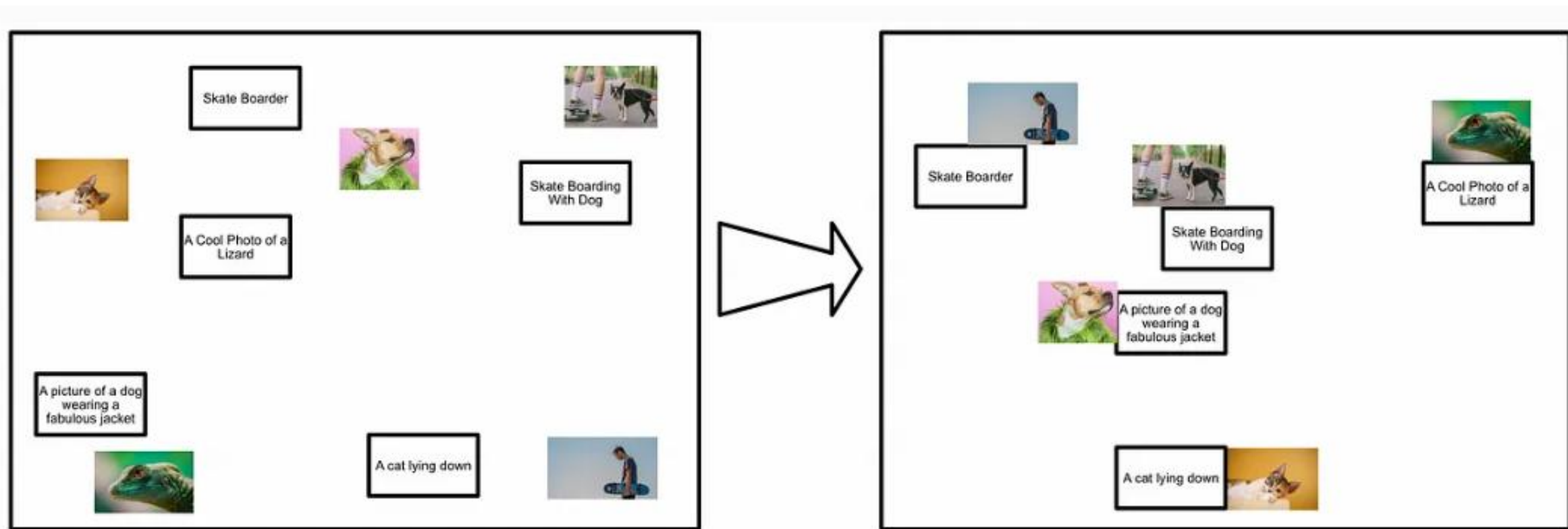
大模型

- DALL-E
- CLIP (Contrastive Language-Image Pre-training)



大模型

- CLIP (Contrastive Language-Image Pre-training)



多模态数据

- 各个模态数据之间的不可替代性
- 语言的特殊性
- 除了图像、声音、语言，其他模态数据比较少



- 一个视觉问答 benchmark 的大部分题目可以在完全不看图的情况下答对
- 当模型看到 What sport is... 这类问题时，高频答案 tennis 往往就是正确答案，完全不需要理解图片内容。
- 2018 年，[Zech 等人](#)在 PLoS Medicine 发表了一项研究：一个训练用来检测肺炎的深度学习模型，实际上学会了识别拍片医院的来源。他们发现，一个仅用医院来源信息做预测的简单模型就能达到 AUC 0.861，而 CNN 能以 99.95% 的准确率识别影像来自哪家医院。背后的机制很清晰：NIH 数据中肺炎患病率为 1.2%，而 Mount Sinai Hospital 为 34.2%，如此大的基线差异让模型可以完全忽略影像内容，只靠识别医院标记就能获得看起来不错的性能。该论文有超过 1400 次引用。

- 2021 年, [DeGrave 等人](#)在 Nature Machine Intelligence 上报告了类似的发现: COVID-19 胸片检测模型约 50% 的准确率来自虚假的相关性 (spurious confounds), 包括影像上的文字标记、患者体位和侧位标记 (laterality markers), 而非肺部病变本身。

- Asadi, O'Sullivan et al. (2026). MIRAGE: The Illusion of Visual Understanding. [arXiv:2603.21687](#)
- DeGrave et al. (2021). AI for radiographic COVID-19 detection selects shortcuts over signal. [Nature Machine Intelligence](#)
- Zech et al. (2018). Variable generalization performance of a deep learning model to detect pneumonia in chest radiographs. [PLoS Medicine](#)
- Banerjee et al. (2023). Shortcuts Causing Bias in Radiology AI. [JACR/PMC](#)
- Yue et al. (2024). MMMU-Pro: A More Robust Multi-discipline Multimodal Understanding Benchmark. [arXiv:2409.02813](#)
- Du et al. (2025). ViLP: Knowledge Exploration using Vision, Language, and Pose. [arXiv:2501.00569](#)
- Bai et al. (2024). Hallucination of Multimodal Large Language Models: A Survey. [arXiv:2404.18930](#)
- Wang et al. (2025). Revisiting Performance Claims for Chest X-Ray Models Using Clinical Context. [arXiv:2509.19671](#)
- Jabbour et al. (2020). Exploiting and Preventing Shortcuts in Chest X-ray. [PMLR](#)
- Rahmanzadehgervi et al. (2024). VLMs are Blind. [arXiv:2407.06581](#)
- Goyal et al. (2017). VQA v2. [visualqa.org](#)
- Agrawal et al. (2018). VQA-CP. [arXiv:1712.00377](#)